Chapter 26: Vision and Optical Instruments

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Lecture 26

Part I

The Human Eye

Outline of Part I: The Human Eye

Outline of Part I: The Human Eye

Vision and Optical Instruments

The eye is the most sophisticated optical instrument ever developed — a self-focusing, self-adjusting, self-repairing lens system that can detect a single photon.

Central Question of This Chapter

- How do optical systems collect light and form images?
- How does the eye fail — and how do we correct it?
- How do microscopes and telescopes extend human vision?
- Why are all real optical systems imperfect?

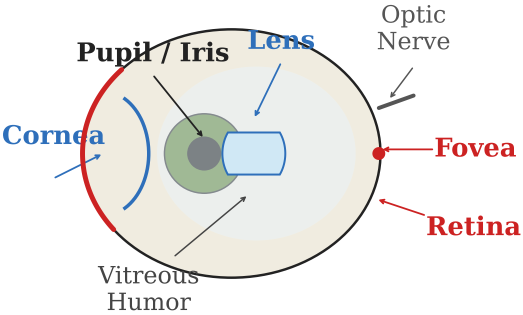
Connecting Thread

Each topic in this chapter answers the same question:

How does an optical system form an image of an object, and what are its limits?

Outline of Part I: The Human Eye

Anatomy of the Eye



Key Structures

- **Cornea** — primary refracting surface (∼2/3 of total power)
- **Lens** — adjustable, provides fine focus
- **Iris/Pupil** — controls light intensity
- **Retina** — light-sensitive detector
- **Fovea** — highest acuity region
- **Optic nerve** — signal pathway to brain

The Eye as a Converging Lens System

Thin-Lens Equation Applied to the Eye

The image distance d_i is **fixed** ($\approx 2.0\,\text{cm}$):

$$\boxed{\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} = P}$$

$P = 1/f$: power in **diopters** (D)

Since d_i is fixed, the eye adjusts f (via P) to focus objects at different distances.

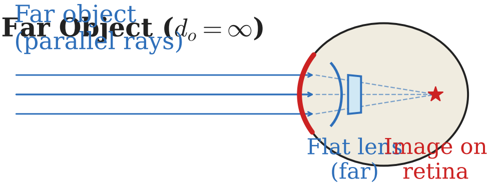
Power of the Eye

- Total at rest: $\approx 60\,\text{D}$
- Cornea: $\approx 43\,\text{D}$ (fixed)
- Lens: $\approx 17\,\text{D}$ (adjustable)
- Maximum: $\approx 67\,\text{D}$

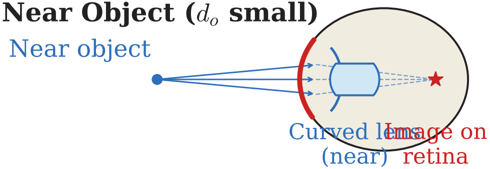
Outline of Part I: The Human Eye

Accommodation: Adjusting Focus

Far Object ($d_o = \infty$)



Near Object (d_o small)



Accommodation

The ciliary muscles **change the shape** of the lens, altering its focal length.

- **Far object:** ciliary muscles relax, lens flattens, f increases (P decreases)
- **Near object:** ciliary muscles contract, lens rounds, f decreases (P increases)

The **image must always land on the retina**. Since d_i is fixed, f must change when d_o changes.

Near Point and Far Point

Far Point

The **farthest** distance at which the eye can form a clear image.

A normal (emmetropic) eye: far point $= \infty$

($d_o = \infty \Rightarrow P = 1/d_i$, minimum power)

Near Point

The **closest** distance at which the eye can form a clear image.

Normal young adult: near point $\approx 25\,\text{cm}$

($d_o = 25\,\text{cm} \Rightarrow P$ maximum)

Accommodation Range

$$\begin{aligned}\Delta P &= P_{\text{max}} - P_{\text{min}} \\ &= \frac{1}{d_{o,\text{near}}} + \frac{1}{d_i} - \frac{1}{d_i} \\ &= \frac{1}{d_{o,\text{near}}}\end{aligned}$$

As we age, the lens hardens and accommodation range decreases — this is **presbyopia**.

The Inverted Retinal Image

The lens of the eye forms a **real, inverted** image on the retina — just as any converging lens forms an inverted real image when $d_o > f$.

Common Misconception

- Students often assume we see the world "right-side up" because the lens rights the image.
- In fact, the retinal image is **inverted** and **reduced**.
- The brain interprets the neural signals from the retina and constructs an upright perception.

Key Insight

Perception is constructed by the brain, not recorded passively by the eye.

The eye is the *detector*; the brain is the *image processor*.

Example: Power of the Eye — (cp2e_ch26_ex1)

Power of the Eye for a Distant Object

Calculate the power of the eye when viewing an object at infinity. Assume $d_i = 2.00\,\text{cm}$.

$$\frac{1}{d_o} + \frac{1}{d_i} = P$$

$$d_o = \infty \Rightarrow \frac{1}{d_o} = 0$$

$$P = \frac{1}{d_i} = \frac{1}{0.0200\,\text{m}}$$

$$\boxed{P = 50.0\,\text{D}}$$

At rest (far point $= \infty$), the eye operates at minimum power: $50.0\,\text{D}$.

The total power $\approx 60\,\text{D}$ includes a corneal contribution of $\approx 43\,\text{D}$ — but the simple model treats the eye as a single thin lens.

- What happens to P as d_o decreases?
- Why can't d_i change to refocus?

Example: Near-Point Power — (cp2e_ch26_ex2)

Maximum Power of the Eye

An eye has a near point of 25.0 cm and $d_i = 2.00\,\mathrm{cm}$. What is its maximum power?

$$P = \frac{1}{d_o} + \frac{1}{d_i}$$

$$P = \frac{1}{0.250\,\mathrm{m}} + \frac{1}{0.0200\,\mathrm{m}}$$

$$P = 4.00\,\mathrm{D} + 50.0\,\mathrm{D}$$

$$\boxed{P = 54.0\,\mathrm{D}}$$

Accommodation range:
$54.0\,\mathrm{D} - 50.0\,\mathrm{D} = 4.0\,\mathrm{D}$.

This is the extra power the lens muscles must supply when focusing at the near point.

- Is $P_{\mathrm{max}} > P_{\mathrm{min}}$? ✓
- Why does the near point move farther away with age?

Check Your Understanding

An object is moved closer to a normal eye. Does the power of the eye increase or decrease? Does the focal length increase or decrease?

Come to lecture to see the answer.

Part II

Vision Correction

Outline of Part II: Vision Correction

Outline of Part II: Vision Correction

Myopia and Hyperopia

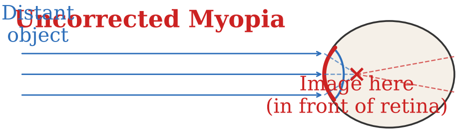
Presbyopia and Laser Correction

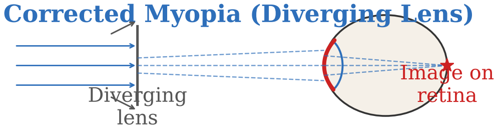
Vision Defects: When Image Formation Fails

When the eye cannot bring an image to focus on the retina — because the eyeball is the wrong size or the lens cannot adjust enough — we need a **corrective lens**.

Condition	**Cause**	**Correction**
Myopia	Eye too long; image falls *in front of* retina	Diverging (concave) lens
Hyperopia	Eye too short; image falls *behind* retina	Converging (convex) lens
Presbyopia	Reduced accommodation range	Reading glasses (converging)

Myopia (Nearsightedness)





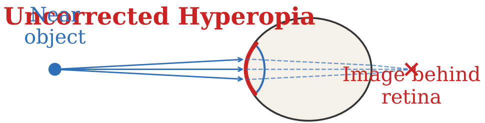
Myopia

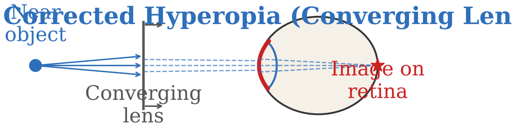
- Far point is *closer* than infinity
- Distant objects focus in front of retina
- Cannot see far objects clearly

Correction

A **diverging lens** makes incoming parallel rays appear to come from the far point — placing the image where the myopic eye *can* focus it.

Hyperopia (Farsightedness)





Hyperopia

- Near point is *farther* than normal
- Near objects focus behind retina
- Cannot see near objects clearly

Correction

A **converging lens** pre-converges the rays from a near object, making them appear to come from the eye's near point or beyond.

Finding the Corrective Lens Power

Strategy for Vision Correction

A corrective lens creates a **virtual image** at the eye's far point (myopia) or near point (hyperopia), so the eye can focus it.

The object is at the *desired viewing distance*; the image is at the *limit of the defective eye*.

Myopia: Correcting Lens

Object at $d_o = \infty$ (distant) must form image at the far point $d_i = -|\text{far point}|$ (virtual, same side as object).

$$P_{\text{lens}} = \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{\infty} - \frac{1}{|\text{far pt}|}$$

Hyperopia: Correcting Lens

Object at $d_o = 25\,\text{cm}$ (near point) must form image at the eye's near point $d_i = -|\text{near pt}|$ (virtual).

$$P_{\text{lens}} = \frac{1}{d_o} - \frac{1}{|\text{near pt}|}$$

Example: Correcting Myopia — (cp2e_ch26_ex3)

Correcting Myopia

A person has a far point of 80.0 cm. What power corrective lens do they need to see distant objects clearly?

$$d_o = \infty, \quad d_i = -0.800\,\mathrm{m}$$

$$P = \frac{1}{d_o} + \frac{1}{d_i}$$

$$P = 0 + \frac{1}{-0.800\,\mathrm{m}}$$

$$\boxed{P = -1.25\,\mathrm{D}}$$

Negative power $\Rightarrow$ **diverging lens** — correct for myopia.

The virtual image at $-80.0\,\mathrm{cm}$ is at the person's far point; the myopic eye can now focus it.

- Why is d_i negative here?
- Does a more negative P correct more or less severe myopia?

Example: Correcting Hyperopia — (cp2e_ch26_ex4)

Correcting Hyperopia

A farsighted person has a near point of 60.0 cm. What power lens allows them to read at 25.0 cm?

$$d_o = 0.250\,\mathrm{m}, \quad d_i = -0.600\,\mathrm{m}$$

$$P = \frac{1}{d_o} + \frac{1}{d_i}$$

$$P = \frac{1}{0.250\,\mathrm{m}} + \frac{1}{-0.600\,\mathrm{m}}$$

$$P = 4.00\,\mathrm{D} - 1.67\,\mathrm{D}$$

$$\boxed{P = 2.33\,\mathrm{D}}$$

Positive power $\Rightarrow$ **converging lens** — correct for hyperopia.

The virtual image at -60.0 cm is at the eye's near point.

- Is the image real or virtual? Why?
- Does the corrective lens replace the eye's focusing or assist it?

Outline of Part II: Vision Correction

Myopia and Hyperopia

Presbyopia and Laser Correction

Presbyopia and Laser Surgery

Presbyopia

- Age-related reduction in accommodation
- Lens hardens and cannot round sufficiently
- Near point recedes beyond comfortable reading distance
- Corrected with **reading glasses** (converging)
- Not the same as hyperopia — the eye is the right size, but the lens cannot deform

Laser Vision Correction

- LASIK reshapes the **cornea**, permanently changing its curvature
- Changes optical power at the source
- Does *not* strengthen muscles or restore accommodation
- Most effective for myopia and mild astigmatism

Laser surgery changes the *geometry* of the cornea — it is optical engineering applied to biological tissue.

Check Your Understanding

A patient's far point is 50 cm. Is this person myopic or hyperopic? What sign of corrective lens power do they need?

Come to lecture to see the answer.

Part III

Color and Color Vision

Outline of Part III: Color and Color Vision

Outline of Part III: Color and Color Vision

The Visible Spectrum

Rods, Cones, and Color Mixing

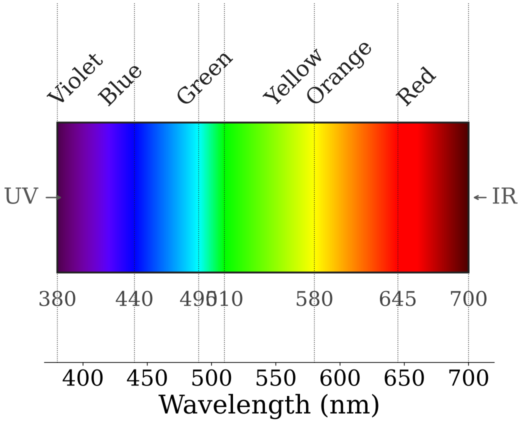
The Visible Spectrum

Visible Range

- $\lambda \approx 380\,\text{nm}$ (violet) to $700\,\text{nm}$ (red)
- A narrow slice of the full EM spectrum
- **Color is a perception**, not a physical property of light
- Same wavelength can appear different colors under different contexts

Key Distinction

Physical: spectral distribution of incoming light
Perceived: brain's interpretation of receptor signals



Outline of Part III: Color and Color Vision

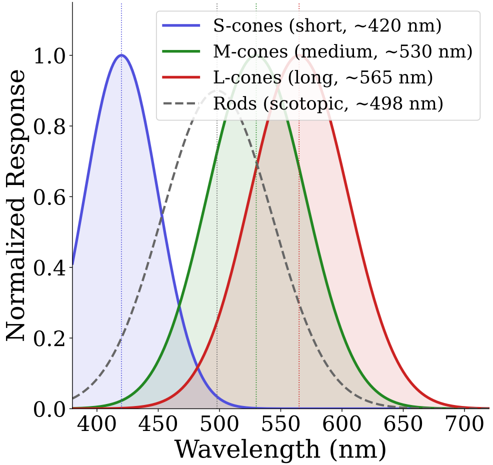
Rods and Cones

Rods

- $\sim$120 million in each eye
- Sensitive to low light levels (night vision)
- No color discrimination
- Absent from fovea; concentrated in periphery

Cones

- $\sim$6 million in each eye
- Require bright light (day vision)
- Color sensitive: three types (S, M, L)
- Concentrated in fovea



Trichromatic Theory of Color Vision

Color perception arises from the **ratio of responses** of three cone types: S (short, $\sim$420 nm), M (medium, $\sim$530 nm), L (long, $\sim$560 nm).

Metamerism

Two *physically different* spectra can produce the *identical perception* if they stimulate the three cone types in the same ratio.

Example: pure yellow light ($\sim$580 nm) and a red-green mixture can appear identical — even though the spectra are completely different.

Practical Applications

- TV/computer displays use only 3 colors (RGB)
- Printing uses CMYK
- Human eye cannot distinguish metamers
- Camera sensors mimic trichromatic detection

Additive vs. Subtractive Color Mixing

Additive (RGB)

Screens, projectors

Emit light

R + G + B $\rightarrow$ white

R + G = Yellow

Subtractive (CMY)

Paints, inks, pigments

Absorb light

C + M + Y $\rightarrow$ black

C absorbs Red

Why different? Screens *emit* light; adding colors adds photons. Pigments *remove* wavelengths; mixing removes more.

Cyan = complement of Red
Magenta = complement of Green
Yellow = complement of Blue

Color Blindness

Color blindness results from the **absence or malfunction** of one or more cone types.

- **Deuteranopia:** missing M-cones (green); red-green confusion — most common ($\sim$5% of men)
- **Protanopia:** missing L-cones (red); red-green confusion
- **Tritanopia:** missing S-cones (blue); rare
- **Monochromacy:** only one cone type; rare

Signal encoding view:

Three cone types $\rightarrow$ 3D color space.
Two cone types $\rightarrow$ 2D color space.
One cone type $\rightarrow$ 1D (no hue).

Color blindness = reduced dimensionality of color encoding.

Color blind individuals lose *one dimension* of the color signal; fewer colors can be distinguished.

Check Your Understanding

Two different light sources produce the same color perception in a normal observer. Does this mean they emit the same wavelengths? Explain.

Come to lecture to see the answer.

Part IV

Microscopes

Outline of Part IV: Microscopes

Outline of Part IV: Microscopes

The Simple Magnifier

A **simple magnifier** (magnifying glass) is a single converging lens used to view objects placed closer than its focal length, producing a **virtual, upright, enlarged** image.

Angular Magnification

The magnifying glass increases the **angular size** of the object as seen by the eye.

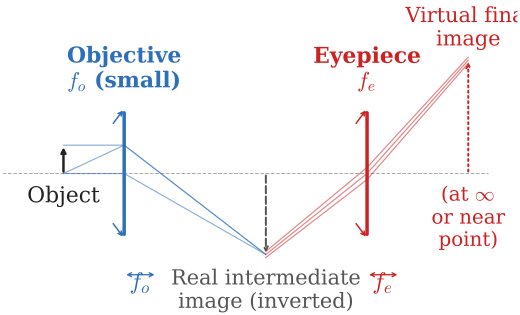
$$M = \frac{\theta'}{\theta} \approx \frac{25\,\mathrm{cm}}{f}$$

where 25 cm is the standard near-point distance and the image is at infinity ($d_o = f$).

- Shorter $f \Rightarrow$ larger M
- A $10\times$ lens has $f \approx 2.5\,\mathrm{cm}$
- Object placed at the focal point: image at ∞, eye relaxed
- Object inside f: virtual, upright, magnified

Outline of Part IV: Microscopes

The Compound Microscope



Two-Lens System

- **Objective lens** (f_o small): creates a real, inverted, *enlarged* intermediate image
- **Eyepiece lens** (f_e): acts as magnifying glass for the intermediate image
- Final image: virtual, inverted, very large

Image Chain

Object $\xrightarrow{\text{objective}}$ real intermediate image $\xrightarrow{\text{eyepiece}}$ virtual final image

Magnification of the Compound Microscope

Total Magnification

The total magnification is the **product** of objective and eyepiece magnifications:

$$\boxed{m = m_o \cdot m_e}$$

Objective Magnification

$$m_o \approx -\frac{d_i}{f_o}$$

d_i = image distance from objective (large)
Negative: intermediate image is inverted

Eyepiece Magnification

$$m_e = \frac{25\,\text{cm}}{f_e}$$

Standard near-point at 25 cm

Example: a microscope with $m_o = 40\times$ objective and $m_e = 10\times$ eyepiece gives total $m = 400\times$.

Bigger $\neq$ More Detailed

Magnification alone does not reveal finer detail. **Resolution** is what determines the smallest visible feature.

Outline of Part IV: Microscopes

Simple Magnifier

The Compound Microscope

Resolution and Numerical Aperture

Resolution vs. Magnification

Empty Magnification

Magnifying beyond the resolution limit produces a **larger blur**.

Bigger image $\neq$ more detail.

Numerical Aperture

$$\boxed{NA = n \sin\alpha}$$

n: index; α: half-angle of light cone at objective

Rayleigh Resolution Limit

$$\delta \approx \frac{0.61\,\lambda}{NA}$$

- Smaller δ = sharper image
- Short λ or large NA improves resolution
- Oil immersion: $n > 1 \Rightarrow$ larger NA

Example: Compound Microscope Magnification — (cp2e_ch26_ex5)

Compound Microscope

$f_o = 5.00\,\text{mm}$, $f_e = 25.0\,\text{mm}$, image distance $d_i = 15.0\,\text{cm}$. Find m.

$$m_o \approx -\frac{d_i}{f_o} = -\frac{150\,\text{mm}}{5.00\,\text{mm}} = -30.0$$

$$m_e = \frac{250\,\text{mm}}{f_e} = \frac{250\,\text{mm}}{25.0\,\text{mm}} = 10.0$$

$$m = m_o \cdot m_e = (-30.0)(10.0)$$

$$\boxed{m = -300}$$

$|m| = 300\times$; negative sign $\Rightarrow$ final image is **inverted**.

- Which lens contributes more magnification?
- What limits how much we can magnify?

Check Your Understanding

A microscope is magnifying at $400\times$ but cannot resolve two features $1\,\mu\text{m}$ apart. Would increasing magnification to $1000\times$ help? Why or why not?

Come to lecture to see the answer.

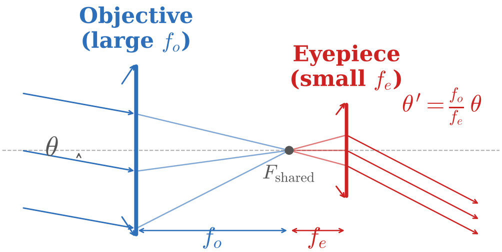
Part V

Telescopes

Outline of Part V: Telescopes

Outline of Part V: Telescopes

The Refracting Telescope



Two Functions

- **Light gathering:** large objective collects more photons (brightness)
- **Angular magnification:** increases apparent angular size

For distant objects: the objective forms an image at its focal point. The eyepiece magnifies this image.

Angular Magnification of a Telescope

Telescope Magnification

$$\boxed{M = \frac{\theta'}{\theta} = -\frac{f_o}{f_e}}$$

f_o: focal length of objective; f_e: focal length of eyepiece; negative sign = inverted image

Design Principles

- Large f_o + small f_e
  $\Rightarrow$ high magnification
- Large objective diameter
  $\Rightarrow$ more light, better resolution
- Long tube length for large f_o

Telescope vs. Microscope

Feature	**Telescope**	**Microscope**
Object dist.	Very large	Very small
Priority	Light collect.	Resolution
Objective	Gathers light	Enlarges image

Outline of Part V: Telescopes

The Refracting Telescope

The Reflecting Telescope

The Reflecting Telescope

Most large modern telescopes use a **concave mirror** as the objective — eliminating chromatic aberration and allowing very large apertures.

Advantages over Refracting

- **No chromatic aberration:** mirrors reflect all wavelengths equally
- **Easier to build large:** mirrors can be supported from behind; lenses must be supported at edges
- **Hubble Space Telescope:** 2.4-m mirror
- **James Webb Space Telescope:** 6.5-m mirror

Why does aperture matter?

Larger aperture $\Rightarrow$ more photons collected per second $\Rightarrow$ fainter objects visible.

Light-gathering power scales as D^2: doubling diameter quadruples light collected.

Example: Telescope Magnification — (cp2e_ch26_ex6)

Telescope Magnification

An astronomical telescope has an objective with $f_o = 1.20\,\text{m}$ and an eyepiece with $f_e = 40.0\,\text{mm}$. Find the angular magnification.

$$M = -\frac{f_o}{f_e}$$

$$M = -\frac{1.20\,\text{m}}{0.0400\,\text{m}}$$

$$\boxed{M = -30.0}$$

The telescope magnifies angular size by $30\times$. The negative sign means the image is **inverted** (acceptable for astronomical use; terrestrial telescopes add an erecting lens).

- How would you increase M without changing f_o?
- Why use a long f_o rather than a large lens power?

Check Your Understanding

Two telescopes have the same magnification but different objective diameters. Which telescope can see fainter stars? Explain.

Come to lecture to see the answer.

Part VI

Aberrations

Outline of Part VI: Aberrations

Outline of Part VI: Aberrations

Types of Aberrations

Why Real Lenses Differ from Ideal Lenses

The thin-lens equation assumes **paraxial rays** (small angles) and a **perfectly spherical** lens of zero thickness. Real lenses violate these assumptions.

Aberrations = Deviations from the Ideal

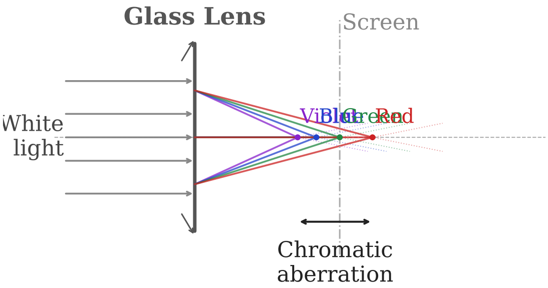
- **Chromatic aberration**: different wavelengths focus at different points
- **Spherical aberration**: marginal vs. paraxial rays focus at different points
- **Coma**: off-axis point sources produce comet-like blur
- **Astigmatism**: tangential vs. sagittal rays focus differently
- **Field curvature**: best-focus surface is curved, not flat

Engineering tradeoff:

Correcting aberrations requires additional lens elements $\Rightarrow$ more cost, weight, and complexity.

Perfect correction is never achieved — only minimized for the specific design purpose.

Chromatic Aberration



Cause

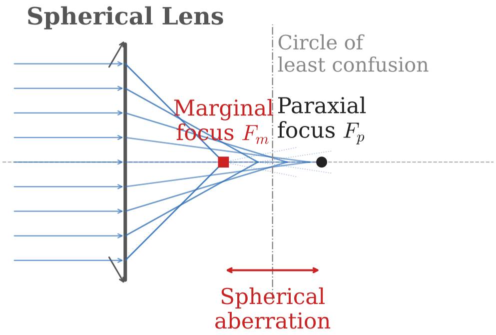
The index of refraction $n(\lambda)$ varies with wavelength (**dispersion**).

n larger for shorter λ (violet) $\Rightarrow$ violet light bends more $\Rightarrow$ shorter focal length

Correction

- **Achromatic doublet:** crown glass + flint glass combine to equalize focal lengths for two wavelengths
- Reflecting telescopes: mirrors have no chromatic aberration

Spherical Aberration



Cause

A *spherical* surface does not focus all parallel rays to the same point.

Marginal (outer) rays refract too strongly; paraxial (central) rays refract correctly.

Correction

- **Aspherical lens surfaces**
- **Stop down** (reduce aperture) — passes only paraxial rays
- **Multiple lens elements** cancel aberrations

Other Aberrations and Correcting Strategies

Coma

Off-axis point sources smear into comet-like tails. Gets worse at larger angles from the axis.

Astigmatism

Rays in different planes focus at different distances. Not to be confused with the eye defect astigmatism (same principle).

Field Curvature

The sharpest image lies on a curved surface, not a flat detector — problematic for cameras.

Correction Strategies

- Use **multiple lens elements** (compound lens)
- Use **aperture stops** to restrict rays
- Use **aspherical surfaces**
- Use **mirrors** (chromatic: eliminated)
- Accept residual aberration appropriate for the application

Modern camera lenses use 10–20 elements to control all five Seidel aberrations simultaneously.

Example: f-Number and Aperture — (cp2e_ch26_ex7)

f-Number of a Camera Lens

A camera lens has a focal length of 50.0 mm and aperture diameter 25.0 mm. Find the f-number.

$$f/\# = \frac{f}{D}$$

$$f/\# = \frac{50.0\,\mathrm{mm}}{25.0\,\mathrm{mm}}$$

$$\boxed{f/\# = 2.0}$$

$f/2$ is a relatively "fast" (wide-aperture) lens — gathers much light, but spherical aberration is significant at full aperture.

Stopping down to $f/8$ reduces aberrations but reduces light.

- Smaller $f/\#$ means *larger* aperture (more light)
- $NA \approx 1/(2 \cdot f/\#)$

Check Your Understanding

A refracting telescope shows a colored halo around a bright star. What type of aberration is this? How would switching to a reflecting telescope help?

Come to lecture to see the answer.

Part VII

Summary

Outline of Part VII: Summary

Chapter Summary

Outline of Part VII: Summary

Chapter Summary

Chapter 26 Summary

Key Equations

Eye power	$P = \frac{1}{d_o} + \frac{1}{d_i}$
Simple magnifier	$M = \frac{25\,\text{cm}}{f}$
Microscope	$m = m_o \cdot m_e$
Telescope	$M = -\frac{f_o}{f_e}$
Num. aperture	$NA = n \sin \alpha$
f-number	$f/\# = f/D$

Conceptual Themes

- Eye fixes d_i; adjusts P via accommodation
- Corrective lenses move virtual images to eye's usable range
- Color is a brain construction, not a physical property
- Magnification $\neq$ resolution
- Aperture determines light gathering and resolution limit
- Aberrations are deviations from the paraxial ideal